Good Data or Leaked Data?

Do Aggregate Privacy Checks Hide Risk to Outlier Records in Synthetic Tabular Data?

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Research Question

Synthetic tabular data is often proposed as a way to share data without exposing real people. Papers that introduce new generators usually summarize privacy with one dataset-level check, such as the distance to the closest record (DCR). Following the proposal feedback, we have narrowed our project from a literature review to a question we test ourselves:

> *When a synthetic dataset passes the aggregate privacy check that generator papers commonly report, are outlier records in the training data still exposed at a much higher rate than typical records?*

We test two hypotheses. **H1:** a simple membership inference attack succeeds more often on outliers than on typical records. **H2:** the aggregate DCR check does not reveal this gap.

Related Work

Tabular data generators. The Synthetic Data Vault [1] models single and multi-table relational data with Gaussian copulas and found that data scientists working with its synthetic data reached conclusions similar to those using the real data. CTGAN [2] adapts GANs to mixed-type tables with mode-specific normalization for multimodal numeric columns and a conditional generator for imbalanced categorical columns; the same paper introduces TVAE and a benchmark in which the VAE is often competitive. TabDDPM [3] combines Gaussian diffusion for numeric features with multinomial diffusion for categorical ones and outperforms GAN and VAE baselines on machine-learning utility. It also shows that SMOTE, which simply interpolates between real records, is a strong utility baseline. We use the copula, CTGAN, TVAE and an interpolation baseline in our experiments.

Utility and fidelity evaluation. Esteban et al. [4] introduced *train on synthetic, test on real* (TSTR) and its reverse (TRTS), now the standard utility tests. CTGAN and TabDDPM evaluate utility the same way, as the accuracy of models trained on synthetic data. Herurkar et al. [5] show that common fidelity scores such as SDV Fidelity can be insensitive to added noise and dropped modes, and propose FAED, FPCAD and RFIS, distribution distances inspired by FID and the Inception Score. Their method is to inject known defects into data and check whether each metric responds; we adapt this design to privacy checks.

Privacy evaluation. Where generator papers check privacy, they compare the training set with held-out data as a whole. Esteban et al. [4] conclude their model does not memorize because reconstruction errors (Kolmogorov–Smirnov test) and an MMD three-sample test do not differ between training and test data; TabDDPM reports DCR to show its samples are less similar to training records than SMOTE's. Membership inference attacks [7] instead ask whether a specific record was in the training data, and Carlini et al. [8] argue that such attacks should be judged by their true-positive rate at a low false-positive rate, since an average-case score can hide that a few records are reliably exposed. Stadler et al. [6] apply shadow-model attacks to synthetic data and find that outlier records stay highly vulnerable; they argue directly against similarity-based privacy metrics. Their attacks train many generative models per target, too costly to be routine in generator papers. Formal protection

comes from differential privacy, through DP-SGD [9] (used by Esteban et al.) or PATE-GAN [10], at a significant cost in utility [6].

Gap. The record-level work shows that outliers are at risk; the generator papers report aggregate checks. No paper we found tests whether those aggregate checks would detect leakage concentrated on outliers. That is our question.

Progress and Preliminary Results

Setup. We use UCI Adult (45,222 complete rows). For each of three seeds we draw a training set (5,000 rows) to fit the generators, a holdout set (5,000) they never see, a test set (10,000) for TSTR, and a reference set (the rest) used only to define outliers: the 10% of records farthest from their 10 nearest reference neighbors. We fit a Gaussian copula, CTGAN and TVAE (`sdv`, 300 epochs), a SMOTE-style interpolation baseline, and a "copy" reference that resamples training rows. Each produces 5,000 rows. We measure *utility* as TSTR F1 with gradient boosting; the *aggregate check* as the DCR share, the fraction of synthetic records closer to a training than a holdout record (0.5 means no sign of copying); and *record-level risk* as the share of training members a distance-based membership attack identifies at a 1% false-positive rate, for typical and outlier records separately [8].

	TSTR F1	DCR share	Members identified at 1% FPR	
Generator		(0.5 = ideal)	Typical	Outliers
Real data (reference)	0.685	–	–	–
Copy (reference)	0.668	1.000	0.635	0.629
Interpolation (SMOTE-style)	0.663	0.929	0.185	0.436
CTGAN	0.599	0.506	0.011	0.009
TVAE	0.586	0.520	0.010	0.013
Gaussian copula	0.083	0.517	0.010	0.017

Table 1: Mean over three seeds. Random guessing identifies 1% of members at 1% FPR.

H1 holds for interpolation only. The interpolation baseline exposes outliers at more than twice the rate of typical records (43.6% vs. 18.5%; Table 1), in all three seeds. CTGAN and TVAE show no measurable leakage with our attack while keeping most of the real-data utility. The copula's low F1 makes it a low-utility reference point.

H2: a controlled test. In this natural experiment the DCR check is not misleading: the leaky baseline fails it clearly (0.93). To test H2 directly we follow Herurkar et al.'s design [5]: starting from TVAE output, we replace a few synthetic rows with near-copies of random outlier training records (numeric features jittered by 5% of a standard deviation). With 50 planted rows (1% of the data), the attack identifies 11.0% of outlier members instead of 1.3%, while the DCR share moves only from 0.520 to 0.524 and no exact duplicates appear. With 250 rows, half of all outliers are exposed and the DCR share (0.545) is still within the range of the unaltered generators (Figure 1). The overall attack rate barely moves (1.0% to 1.4%). A dataset can therefore pass the reported check while exposing many of its outliers.

Plan for the Rest of the Semester

1. **Robustness (next 2 weeks).** Repeat with a second outlier definition (isolation forest) and a 5% cut-off; add a stronger attack that uses several nearest synthetic neighbors; add a second dataset

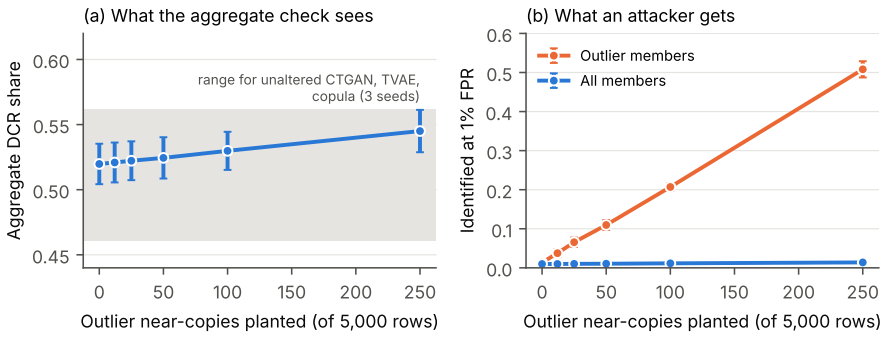


Figure 1: Planting near-copies of outlier training records into TVAE output (mean ± 1 s.d., three seeds). (a) The DCR share stays within the range of unaltered generators (gray). (b) The share of outlier members identified rises steadily; the overall rate stays near 1%.

if time allows.

2. **Final report (following 2 weeks).** Complete the related work, write a reporting checklist that includes a record-level check on outliers, and discuss limits: one dataset, a simple attack, and a planted test that shows what the check *can* miss, not how often real generators leak this way.

All code and results are in our repository (`synthetic_privacy_project/`) and reproduce in about ten minutes on a laptop.

Division of Work

So far. *Austin* reformulated the research question after the proposal feedback, implemented the experiment pipeline (data splits, generators, utility and privacy measures, planted-leakage test), ran the experiments and drafted this update. *Megha* [Megha: describe your contributions so far, e.g., proposal sections, paper reading, related work]. Both co-wrote the proposal.

Remaining. *Austin* will lead the robustness experiments and the results section. *Megha* will lead the privacy and differential-privacy discussion, the reporting checklist, and the limitations section. Both will verify the related work against the full papers and edit the final report.

References

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